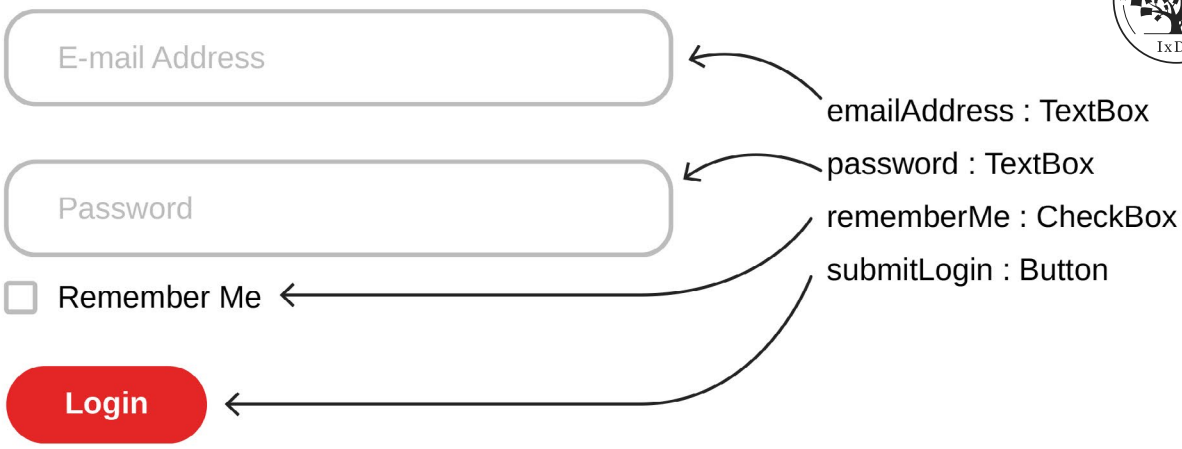

Standardize Your Object-Oriented UI Models

UI Elements for Your UML Software

When you're ready to move from what users interact with to how those interactions actually happen, this template is your shortcut. In the **object-oriented UI (user interface) design** process, this template helps you add standard, clearly named UI elements, like buttons, menus, or checkboxes, directly into your **Unified Modeling Language (UML)** models.

Standardized UI elements keep every diagram consistent and make your object-oriented UI design models easier to understand and implement. It's one of those small but powerful steps that turns your design logic into something ready for real-world development.

How To Use This Template

Use this template once you've built your conceptual model and defined your stories of use. At this stage, you already know which objects users will interact with and what tasks they'll perform. Now it's time to show how they'll do it.

Follow the **step-by-step guide below** to get to grips with the import process for the included **Java file** and start using the UI elements directly in your models. You'll quickly see how they make your object-oriented UI design work clearer, faster, and easier to implement.

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Use Standard User Interface Elements in Your Models

In the object-oriented UI design process, as you create detailed user interface models, you often need to show **which type of UI element** you will use for a particular **attribute**. If you define these elements manually in every model, it takes time and can lead to inconsistencies.

To simplify this process, a **Java file** is included with this guide that you can import into most **Unified Modeling Language (UML) modeling tools**. This file defines a set of typical user interface components, allowing you to reference them directly when specifying attributes in your models. For example:

- **emailAddress : TextBox**

In this example, TextBox is a class defined in the Java file.

The list of elements included in the file is based on standard component names from **Google's developer documentation**. You can modify or extend the file to suit your project's requirements or naming conventions. Many development teams maintain their own list of standard UI components, and you can easily adapt this file using any text editor.

The UI elements currently defined are:

Menu	**MenuBar**	**NavMenu**	**Toolbar**	**Button**
Icon	**Tab**	**TextBox**	**ListBox**	**ComboBox**
SpinBox	**CheckBox**	**RadioButton**	**ExpanderArrow**	**Toggle**

You can add additional components or rename existing ones to match your project's conventions.

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How to Use the Java File in a UML Class Model

Most UML modeling tools allow you to **convert or import Java code** into UML class models. The name of this feature varies depending on the tool. You may find it listed as:

- **Import Java (Astah UML)**
- **Java to UML (Eclipse IDE)**
- **Reverse Java Code (Visual Paradigm)**

Some tools require a commercial (rather than community) license to import Java code.

All UI elements in the file belong to a parent class called **UI**. Depending on how your tool organizes class names, you may see them:

- **Listed under the parent UI class.**
- **Displayed as nested class names** (for example, ComboBox – UI).

After you import the file, you can use these elements directly in your UML models to identify the specific UI components associated with each attribute.

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Learn More. Change Your Life

Learn how to use this template to its full potential in our online course **Object-Oriented UI Design: Build Interfaces Users Love.** Sign up for it today and learn how to create effective personas from scratch if you're not already enrolled.

Intermediate Course

In this course, you will:

- **Get excited** to transform messy requirements into smooth, user-centered solutions people instantly love. **Most designers have never even heard of object-oriented UI design, the new standard in UX.** This gives you an edge. Research shows that clear, intuitive UI **can double conversion rates**. Object-oriented UI design makes that easy.
- **Make yourself invaluable** by keeping design and development in sync. You'll discover how to avoid miscommunication, impractical proposals, and expensive redesigns, and get **recognition** as the person everyone can **trust to drive real impact with critical projects.**
- **Gain confidence and credibility** as the designer who cuts through complexity and delivers clarity. You'll master how to analyze needs, prevent feature creep, and create interfaces that **stay consistent, intuitive, and scalable.**
- **Craft your personal portfolio** with results that prove your value. You'll create career assets like conceptual models and design maps that show how you can move from requirements to **implementation-ready UIs that users love.**

Learn more about this course >

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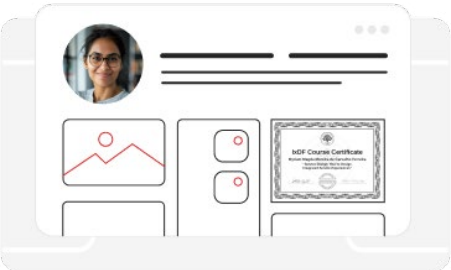

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